
DS 3001 Final Project Paper

Henry Allen ^{*1} Kyra Lim ^{*1} Emma Wunderly ^{*1}

Abstract

Our Project explores refugee migration rates and how we can predict such rates based on origin country and asylum country freedom indexes.

1. Data

The data in our project consists of refugee migration statistics from the **United Nations High Commission for Refugees (UNHCR)**, **Freedom Indexes from the Freedom House (FIW)**, and the **UCDP Priority Conflict Data**. With this data, we plan to investigate refugee rates from countries around the world, and eventually create a model that predicts migration rates based on the political freedom and conflict intensity of the home and host countries.

1.1. Data Sources

FREEDOM IN THE WORLD (FIW) INDEX

The Freedom in the World dataset, published annually by Freedom House, provides scores and ratings from 2013 to 2021 for countries and territories. The data include measures such as political rights ratings, civil liberties ratings, and aggregate scores for subcategories like electoral process, political pluralism and participation, rule of law, and freedom of expression and belief. Each country-year entry also includes an overall status of Free, Partly Free, or Not Free. These variables allow us to capture the political and civil environment in both origin and asylum countries.

UCDP PRIORITY CONFLICT DATA

To better capture the extreme push factors driving displacement, we integrated the `UcdpPrioConflict_v25_1.csv` dataset, which records annual data on organized state-based conflict.

^{*}Equal contribution ¹School of Data Science, University of Virginia, Charlottesville, VA, United States of America. Correspondence to: Henry Allen <jpg7hy@virginia.edu>, Kyra Lim <smy2qe@virginia.edu>, Emma Wunderly <tyw3nq@virginia.edu>.

This data provides the `intensity_level` (1 for minor conflict, 2 for war), allowing us to quantify the severity of active violence. The UCDP data is primarily used to generate conflict metrics for the asylum country, such as total annual and cumulative conflict intensity, serving as critical predictors of migration flow.

UNHCR PERSONS OF CONCERN DATA

The UNHCR Persons of Concern dataset contains annual records beginning in 2004, organized by year, country of origin, and country of asylum. For each country pair, the dataset reports counts of various displaced groups, including refugees, asylum-seekers, internally displaced persons (IDPs), stateless persons, and others of concern. These values reflect the scale of displacement across borders and within states. These values provide the **dependent variable** in our analysis, which is the amount of refugee movement from one country to another in a given year.

The unit of analysis in our study is the `**country-pair-year observation**`, which links refugee flows from an origin country to an asylum country in a given year with their corresponding freedom index values and conflict intensity measures. This allows us to test hypotheses about how political repression, democratic governance, civil liberties, and active conflict shape refugee migration. While the UNHCR data captures migration outcomes, the Freedom House and UCDP data provide measures of the political and violence conditions that act as push and pull factors.

These datasets do consist limitations, such that refugee statistics often underrepresented the quantities of undocumented displacement, and the Freedom House scores, while widely used, are based partly on subjective assessments. Additionally, the UCDP data focuses only on organized, state-based violence. Despite this, their examination provides a unique opportunity to connect quantitative migration patterns with qualitative assessments of political freedom and conflict.

1.2. Data Cleaning

Data cleaning was performed across all three datasets to ensure readiness for integration and analysis. The `**UNHCR Persons of Concern**` data required only initial inspection, as its integrity was confirmed. For the `**Freedom in the World (FIW)**` data, preparation

involved loading with specified sheet/header, **renaming columns** for clarity (e.g., Total to Freedom Total Score), dropping a descriptive row, and **converting relevant columns** to numeric types. The **UCDP Priority Conflict Data** required the most specialized preparation, involving **data aggregation** by country and year to sum the `intensity_level` and generate country-year level features (e.g., `asylum_conflict_total_conflict_intensity`) for the Random Forest Regressor model.

1.3. Data Dictionary

Table 1: Refugee and Related Variables
(`persons_of_concern.csv`)

Column	Description	Predictive Relevance
Refugees	Number of people recognized as refugees under international mandate.	TARGET VARIABLE: The primary focus of prediction.
Asylum-seekers	People seeking international protection.	Key: Highly correlated with refugees, a strong predictor and feature (Importance: 0.156).
IDPs, Stateless, Others of concern	Population counts of other displacement categories.	Strong: Displacement variables that capture overall instability.
Country of Asylum / Origin	Countries involved in displacement flows.	Useful for geographic controls.
ISO Codes	ISO 3166-1 alpha-3 identifiers for merging.	Necessary for joining all datasets.
Host community	Host population size.	Minor contextual control variable.

Table 2: Conflict Variables
(`UcdpPrioConflict_v25_1.csv` and Derived)

Column	Description	Predictive Relevance
<code>intensity_level</code>	The year's intensity of the conflict (1=Minor, 2=War).	Key: Captures the severity of active conflict.
<code>cumulative_intensity</code>	Sum of annual conflict intensity levels.	Captures the total duration and severity of a conflict over time.
<code>asylum_conflict_total_conflict_intensity</code>	Total conflict intensity in the country of asylum.	Key: Top feature (Importance: 0.178), reflecting risk in host country.
<code>asylum_conflict_total_cumulative_intensity</code>	Total cumulative conflict intensity in the country of asylum.	Key: Important predictor (Importance: 0.128), reflecting long-term host country stability.

Table 3: Freedom Variables
(`All_data_FIW_2013-2021.xlsx`)

Column	Description	Predictive Relevance
<code>Freedom_Score_Difference</code>	The difference in FIW Total scores between the Country of Origin and Asylum.	Key: Highly important feature (Importance: 0.107), capturing the 'push' vs. 'pull' factor.
Total (FIW Score)	Aggregate score for all categories (Political Rights + Civil Liberties).	Strong: Holistic measure of governance and human rights in the country.
PR rating / CL rating	Political Rights and Civil Liberties ratings (1-7).	Strong: Coarse-grained summary of governance and rights.
Status (F/PF/NF)	Freedom classification (Free, Partly Free, Not Free).	Strong: Categorical summary of governance, highly relevant to displacement.
PR, CL, A-G Aggregates	Aggregate sub-scores for all categories (e.g., A. Electoral Process, F. Rule of Law).	Strong: Provide a more granular view of the specific drivers of low freedom scores.
A1-G4 Subscores	Individual sub-scores (e.g., A1. Head of government/legislative selection).	Informative but generally collinear with the PR/CL aggregates.

Table 4: Contextual Variables

Column	Description	Predictive Relevance
Year / Edition	Year of observation and report edition.	Useful for trends/time-series effects.
Region	Geographic region.	Captures regional patterns in governance and displacement.
C/T	Country vs. territory indicator.	Minor structural variable.

2. Methods and Results

2.1. Method Overview

The primary objective of this project is to create a robust predictive model that quantifies the relationship between political/civil liberties, active conflict, and refugee migration flows. We frame our task as a **regression problem**, where the dependent variable is the count of refugees from a given origin country to an asylum country in a specific year. The core methodology involves three phases: feature engineering and data integration, predictive model training (Random Forest Regressor), and model validation and assessment. Given the complex, non-linear, and multi-faceted nature of conflict and political stability affecting human migration, we employ a non-parametric approach using a diverse set of features.

The model integrates data from three primary sources:

1. **UNHCR Persons of Concern** (`persons_of_concern.csv`): Provides the target variable, **Refugees**.
2. **Freedom in the World (FIW) Index** (`All_data.FIW_2013-2021.xlsx`): Provides political/civil liberties scores.
3. **UCDP Priority Conflict Data** (`UcdpPrioConflict_v25-1.csv`): Provides annual conflict intensity and cumulative intensity data.

To create the final analytic dataset of approximately 99,946 unique country-pair-year observations (2013–2021), we performed a series of feature engineering and integration steps:

1. **FIW Integration (Push and Pull Factors):** We joined the FIW scores twice to capture the **Origin Country's** score (F_{Origin}) as a **push factor** (low freedom score predicts high migration) and the **Asylum Country's** score (F_{Asylum}) as a **pull factor** (high freedom score predicts high immigration).
2. **Conflict Feature Generation:** The UCDP data was aggregated and merged to derive conflict indicators primarily for the Asylum country, hypothesized to diminish its pull factor, specifically the annual sum of active conflict intensity (`asylum_conflict_total_conflict_intensity`) and the total sum of historical conflict intensity (`asylum_conflict_total_cumulative_intensity`).
3. **Differential Feature Creation:** A crucial engineered characteristic, **Freedom Score Difference** ($F_{\text{Origin}} - F_{\text{Asylum}}$), was calculated to holistically capture the political disparity between the two countries, which is hypothesized to be a strong determinant of the flow magnitude.
4. **Target Transformation:** The target variable, **Refugees**, is highly biased. To achieve a more normal distribution and stabilize the variance of the regression model, we applied a logarithmic transformation: $Y = \log(1 + \text{Refugees})$. The final data set engineered consists of the full set of characteristics (FIW scores, conflict metrics, difference scores, and other contextual variables such as asylum seekers) and the target variable Y .

2.2. Model Justification: Random Forest Regressor

We selected a **Random Forest regression (RFR)** as our predictive model. A sample model can be seen in the figure below.

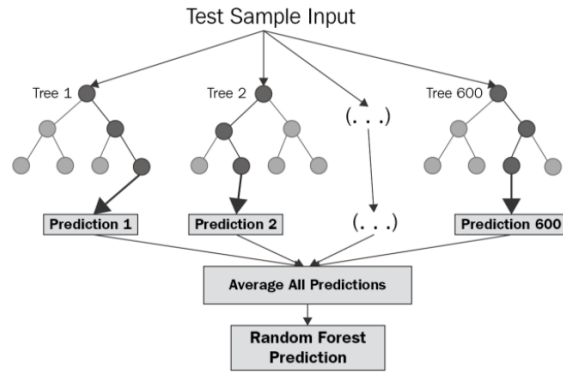


Figure 1: Sample Random Forest Regression Model

This choice is justified by its suitability for the complex and high-dimensional nature of our features:

1. **Handling Non-linearity and Interactions:** The RFR, an ensemble method based on decision trees, can inherently capture highly non-linear relationships and intricate interaction effects among the diverse features (e.g., how the effect of **conflict intensity** in the asylum country is conditional on the **origin country's** freedom score). This avoids the need for manual specification of complex interaction terms.
2. **Robustness to Mixed Data:** RFR is robust against outliers and effectively handles a mix of continuous, categorical, and count data. This is particularly beneficial when combining disparate features like raw population counts (asylum seekers), ordinal ratings (FIW scores), and aggregated intensity metrics (UCDP conflict).
3. **Feature Importance and Interpretability:** The RFR provides a quantitative measure of **feature importance**, which is critical to the interpretability of the model. This allows us to definitively rank and assess the influence of **conflict intensity**, **political freedom differences**, and traditional **asylum-seeker counts** on migration rates. The significance analysis revealed that the main drivers were `asylum_conflict_total_conflict_intensity` (0.178), `asylum seekers` (0.156) and `asylum_conflict_total_cumulative_intensity` (0.128).

2.3. Model Training Procedure

DATA SPLITTING

The engineered dataset was partitioned into training and testing sets using an 80/20 random split. The 80% training set was used to fit the RFR model, while the 20% held-out test set was reserved exclusively for final, unbiased evaluation of the model's performance on unseen data. A random split was chosen over a strict temporal split to ensure maximum diversity of country-year observations in both sets, which helps the model generalize across various geopolitical conditions and time points.

HYPERPARAMETER OPTIMIZATION

To optimize the model's generalization performance and mitigate overfitting, we performed **cross-validated hyperparameter tuning**. We utilized a 5-fold cross-validation and a Randomized Search to efficiently explore the parameter space. Key hyperparameters optimized included:

- **n_estimators**: The number of decision trees in the forest.
- **max_depth**: The maximum depth of each individual decision tree.
- **min_samples_split**: The minimum number of samples required to split an internal node.

The model's performance during cross-validation was evaluated using the negative Mean Squared Error (MSE) on the log-transformed target, with the objective of selecting the parameter combination that minimizes MSE.

2.4. Model Validation Plan

The trained Random Forest Regressor will be evaluated on the withheld test set using both statistical metrics (on the log-scale) and domain-specific metrics (on the original count scale).

STATISTICAL METRICS (LOG SCALE)

The primary statistical metrics calculated on the log-transformed target variable (Y) and its prediction ($\hat{Y}$) are:

- **R² Score**: Measures the proportion of variance in the log-transformed refugee counts that is predictable from the features.
- **Root Mean Squared Error (RMSE)**: Measures the magnitude of the errors on the log scale.

DOMAIN-SPECIFIC METRICS (ORIGINAL SCALE)

To ensure the model's predictions are interpretable in the context of real-world refugee counts, the predictions $\hat{Y}$ will

be inverse-transformed back to the original scale using the formula:

$$\text{Predicted Refugees} = \exp(\hat{Y}) - 1$$

The model's final performance, relevant for practical application, will be assessed using:

- **Mean Absolute Error (MAE)**: Provides the average magnitude of error in terms of raw refugee counts.
- **Mean Absolute Percentage Error (MAPE)**: Provides a percentage error metric, though this can be sensitive to low actual count values.

2.5. Results

To assess how well the Random Forest model predicts refugee movements, we examined its predictions alongside several diagnostic plots. These results reveal both the strengths of the model and the challenges posed by extreme or zero-inflated refugee data.

The below plot highlights the uneven distribution of refugee counts in the dataset, with a large concentration of points at zero. The model tends to slightly over-predict refugee movement in cases where the actual count is zero, which is typical in zero-inflated datasets where most country-pairs experience no movement in any given year. For higher values, the model shows substantial spread around the 45-degree reference line, indicating inconsistent accuracy. Notably, the model appears to under-predict large refugee flows, with many points falling below the line when actual values are high. This idea reflects the tendency of Random Forests to smooth out extreme values and struggle with rare, high-magnitude events.

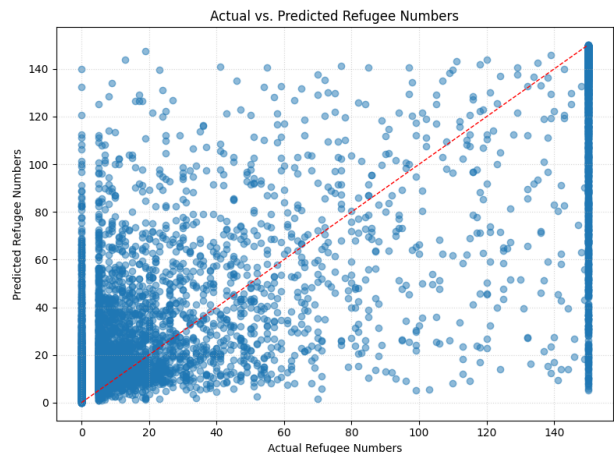


Figure 2: Actual vs. Predicted Refugee Counts (Linear Scale)

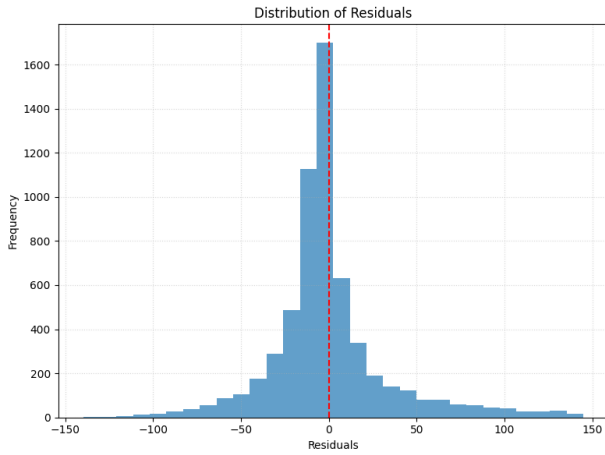


Figure 3: Distribution of Residuals

In the above Figure 3, the distribution of residuals shows that the Random Forest Regressor's errors are centered closely around zero, indicating that the model does not systematically over or under-predict refugee flows on average. The histogram is approximately symmetric, although it exhibits long tails in both directions. These tails reflect that, while most predictions are fairly accurate, the model occasionally makes large errors, mostly for rare or extreme refugee movement events. This behavior is expected in migration and conflict-related datasets, which often contain sudden spikes associated with geopolitical events. Overall, the residual histogram suggests that the model performs reasonably well but struggles to capture more volatile refugee flows.

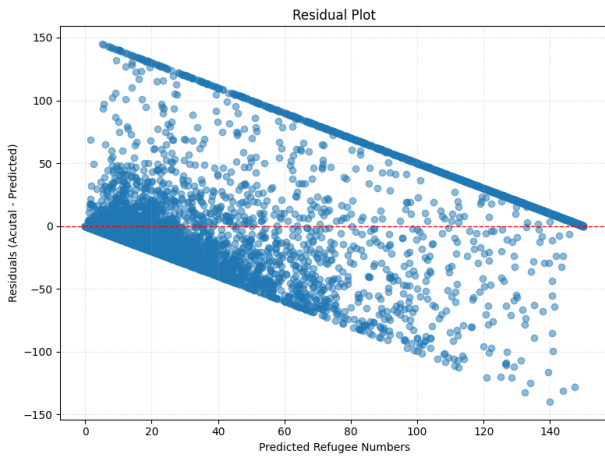


Figure 4: Predicted vs. Residuals Plot

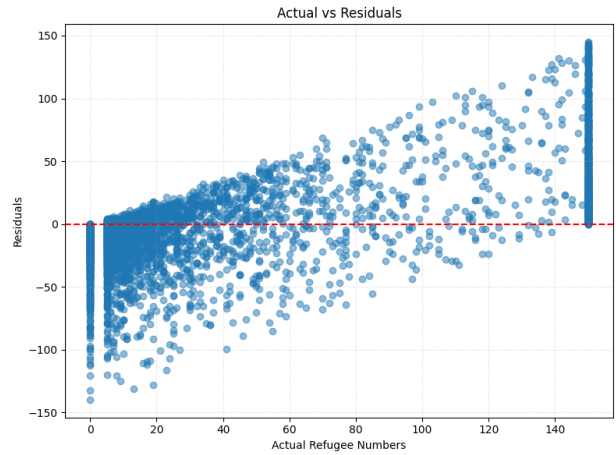


Figure 5: Actual vs. Residuals Plot

Figures 4 and 5 represent the model's residuals against both predicted and actual refugee counts, respectively. This helps illustrate how error patterns vary across the full range of values. The plots reveal a widening spread of residuals as predictions increase, indicating that uncertainty grows when the model attempts to estimate higher refugee flows. The funnel-shaped pattern suggests that while the model remains stable for low and moderate predictions, it becomes increasingly inconsistent as it encounters larger and less frequency displacement events.

When examined against actual values, the residuals show a similar trend. Small and zero flows, which make up a majority of the dataset, are predicted reasonably well, with errors remaining close to zero. In contrast, large displacement events produce substantially larger residuals and are consistently under predicted, highlighting the model's tendency to smooth extreme outcomes toward the center. This behavior reflects the inherent difficulty of modeling rare, conflict-driven surges in refugee movement, where dramatic spikes may occur that are hard for data-driven methods, such as our model, to anticipate. Together, these results show the limitations of tree-based models when applied to highly skewed, zero-inflated outcomes.

Lastly, in the below plot of the log-scaled Actual versus Predicted plot, a clearer view of model performance across the full range of refugee counts can be seen. When scaling the data, the extreme values are compressed, separating the dense cluster of zero and low-value cases. On the log scale, mid-range events exhibit a more coherent linear relationship, indicating that the model captures moderate refugee movements relatively well. However, the highest values still show a significant downward deviation from the 45-degree line, reinforcing that the model underestimates the largest displacement events. The column of points at zero on the log scale emphasizes the heavily zero-inflated nature of the dataset, and the model's tendency to slightly over-predict in

these cases. Overall, the log-scale plot highlights both the strengths and limitations of our model.

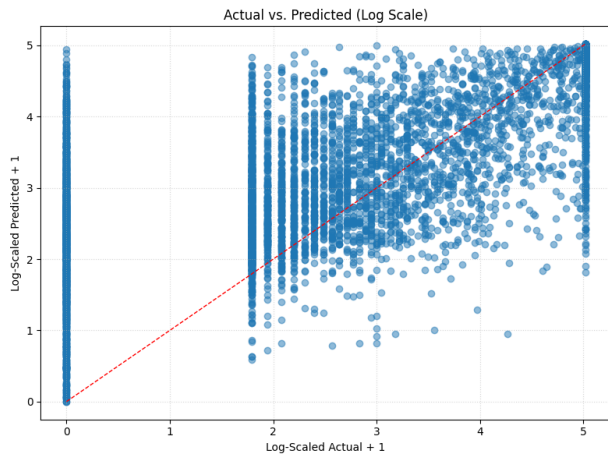


Figure 6: Actual vs. Predicted (Log Scale)

3. Conclusion

Impact Statement

References

Freedom House. Freedom in the world. Technical report, Freedom House, Washington, DC, 2025. URL <https://freedomhouse.org/report/freedom-world>.

UNHCR. Refugee statistics – operational data portal. Technical report, United Nations High Commissioner for Refugees, Geneva, Switzerland, 2025. URL <https://www.unhcr.org>.

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A. Appendix

The `\onecolumn` command above can be kept in place if you prefer a one-column appendix, or can be removed if you prefer a two-column appendix. Apart from this possible change, the style (font size, spacing, margins, page numbering, etc.) should be kept the same as the main body.